Project 1

Site: Eduvos LMS
Course: Natural Language Processing with Python Assessments
Book: Project 1
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Table of contents

1. Project
2. Instructions to Students
3. Section A
4. Terms of AI use for this Assessment
5. Recommended Use for AI in your Assessments
6. Question 1
7. Question 2
8. Question 3
9. Question 4
10. How AI use Affects Marking

Faculty:	Information Technology
Module Code:	ITNPA4-34
Module Name:	Natural Language Processing with Python
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Total Marks:	100
Submission Week:	Block 4 Week 6

This module is presented on NQF level 8.

5% will be deducted from the student's project mark for each calendar day the project is submitted late, up to a maximum of three calendar days. The penalty will be based on the official campus submission date.

Projects submitted later than three calendar days after the deadline or not submitted will get 0%. [1]

This is an individual project

This Project contributes 40% towards the final mark.

[1] Under no circumstances will projects be accepted for marking after the projects of other students have been marked and returned to the students.

Instructions to Students

1. Please ensure that you use the Assignment Template when preparing your submission (where applicable).
2. Please ensure that your answer file (where applicable) is named as follows before submission: **Module Code – Assessment Type – Campus Name – Student Number.**
3. Remember to keep a copy of all submitted assignments.
4. All work must be typed.
5. Please note that you will be evaluated on your writing skills in all your assignments.
6. All work must be submitted through Turnitin. The full originality report will be automatically generated and available for the lecturer to assess. Negative marking will be applied if you are found guilty of plagiarism, poor writing skills, or if you have applied incorrect or insufficient referencing. (See the "instructions to students" book activity before this activity where the application of negative marking is explained.)



Section A

Learning Objective

To test students' abilities to perform foundational text mining and natural language processing.

Project Topic

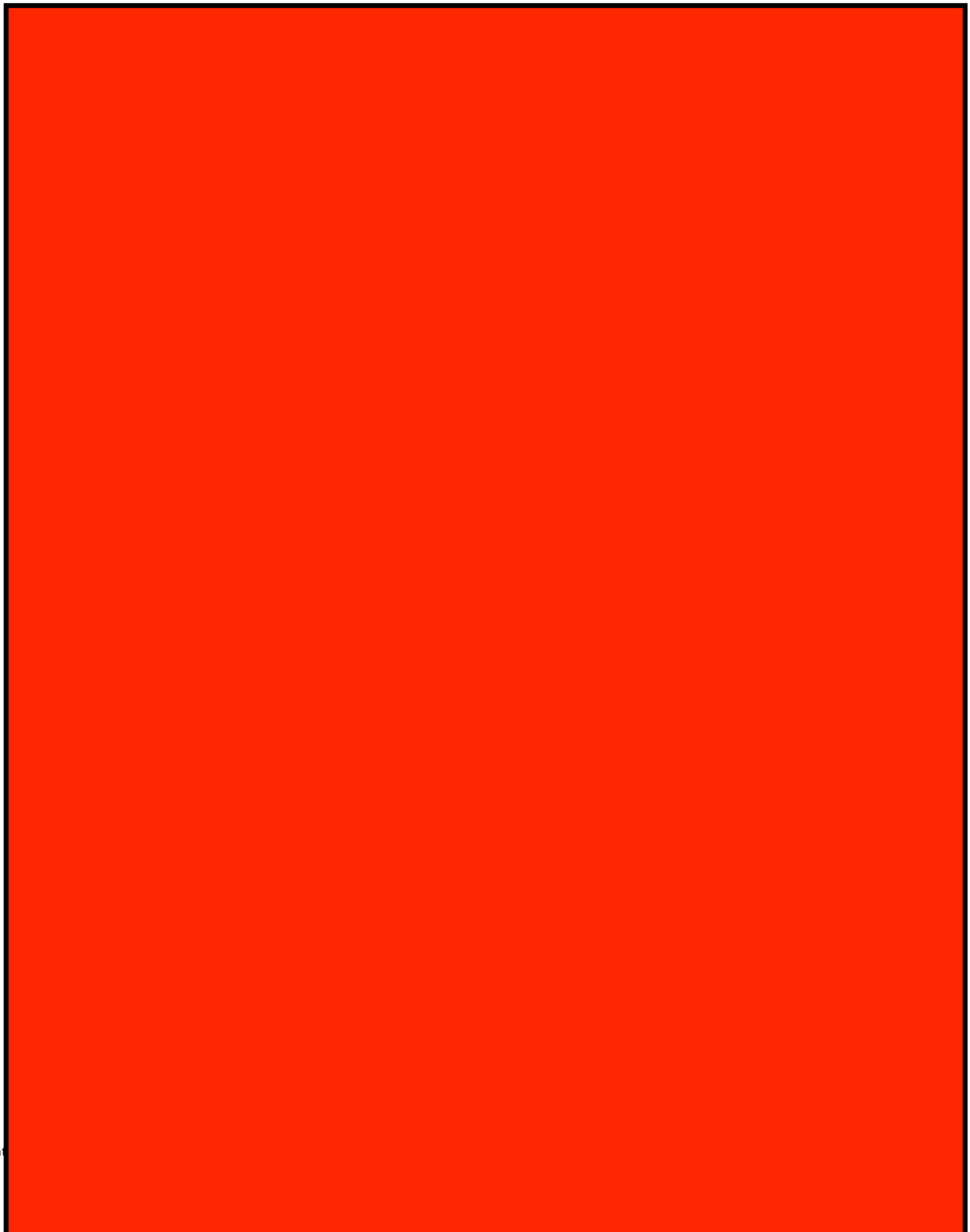
Python programming, text preprocessing, text modelling, text summarisation and analysis, text classification, analytics basic deployment

Scope

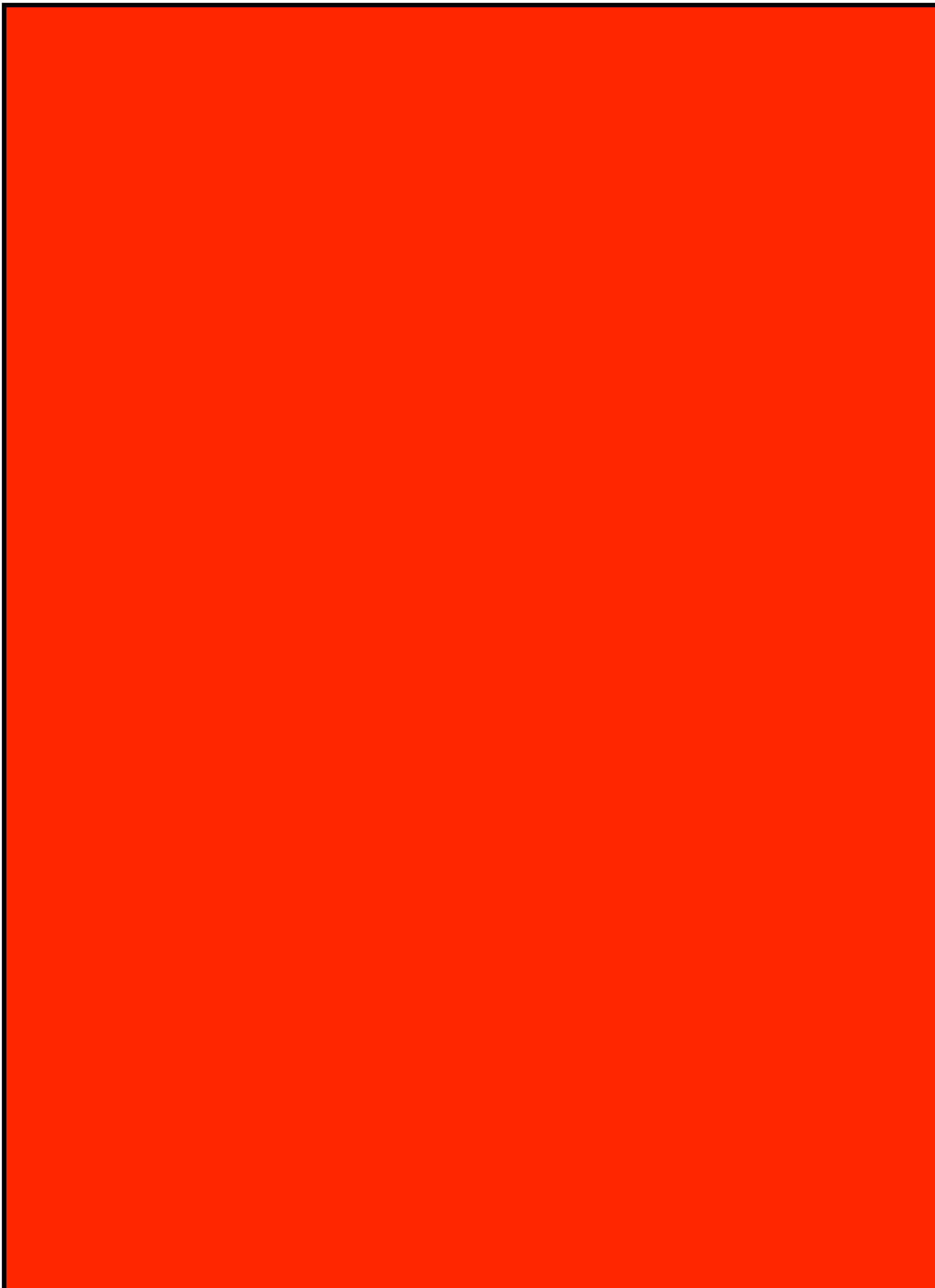
Python programming, text preprocessing, text modelling, text summarisation and analysis, text classification, analytics basic deployment

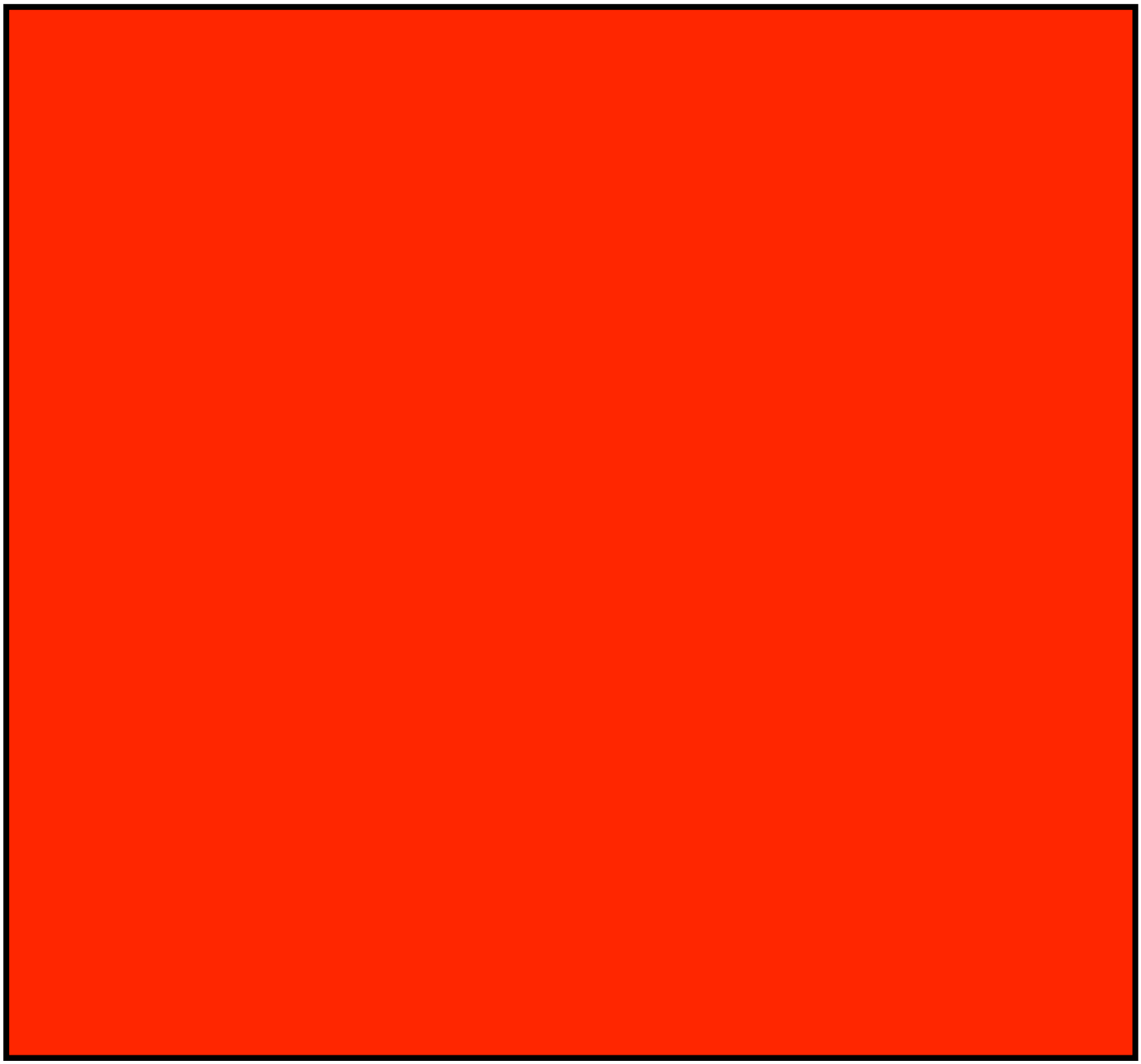
Technical Aspects

Python programming, text preprocessing, text modelling, text summarisation and analysis, text classification, analytics basic deployment



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Research Papers – Expert System

Researchly is a data science research analytics service launched by SAnalytics to help researchers quickly analyse research papers and minimise the screening phase of relevant research articles for one's purpose. Given a database of research articles, research will typically enter keywords that the computer search within a document and output only the relevant work. The company has decided to heighten this capability by working on machine learning inspired approaches that will classify documents semantically based on context. You have task to delve into this space and have been assigned a research category dataset to showcase your skills ('**research_abstracts.json'**) in text mining relevant to the problem in hand.

Source: MATANGA NY (2026)

1.1. Load the data frame into Python and preprocess the datasets by removing all irrelevant information using text denoising techniques.

(6 marks)

1.2. Identify data columns relevant for feature engineering using feature modelling techniques of your choice.

(5 marks)

1.3. Build a baseline model, provide a performance report for model learning and generalization and comment on the performance of your model.

(9 marks)

[Sub Total 20 Marks]

2.1 From the data frame in question 1, train at least four machine learning models of your choice with the goal of obtaining better predictive performance than the baseline model.

(10 marks)

2.2. Besides the use of different models, provide one additional predictive performance enhancement approach of your choice, showcasing with implementation how it improves the design performance

(10 marks)

2.3. Deploy the best performing algorithmic configuration of your design in a Web-interface for testing as in the image below.

Researchly Research Abstract Classifier

This app classifies research abstracts as either AI-related or classical research.

Enter research abstract:

Artificial intelligence (AI) has become a transformative technology in healthcare by enabling more accurate diagnosis, predictive analytics, and personalised treatment planning. Machine learning algorithms can analyse large volumes of medical data to identify patterns that assist clinicians in detecting diseases at earlier stages. In addition, AI-powered decision support systems improve clinical workflows by reducing manual effort and enhancing diagnostic consistency. Despite these benefits, challenges such as data privacy, model interpretability, and ethical considerations remain significant barriers to widespread adoption. Continued research is required to develop robust, transparent, and trustworthy AI systems that can be safely integrated into modern healthcare environmen.

Classify Abstract

Prediction

The abstract is classified as: artificial intelligence

(10 marks)

[30 marks]

Topic Analysis – Research Document Mining



Search engines index thousands of new research publications each year, making it increasingly challenging for researchers and analysts to identify emerging trends and monitor the evolution of research topics. To address this challenge, you have been tasked with designing a prototype system for tracking research trends in the field of artificial intelligence. As a proof of concept, the system will apply topic modelling techniques to a small collection of recent AI research articles provided in the **research_corpus.zip** dataset, with the aim of uncovering the main themes and research directions represented in the corpus.

Source: MATANGA NY (2026)

3.1. Generate a dataframe of research articles including all articles and their respective content.

(5 marks)

3.2. Preprocess each research article content using state-of-the art denoising techniques.

(6 marks)

3.3. Summarize the information content in the first article using a summarization technique of your choice including visualization.

(5 marks)

3.4. Summarise as an aggrate the remaining articles in the corpus using another summarization technique of your choice including appropriate visualization methods

(9 marks)

[Sub Total 25 Marks]

4.1. Using Latent Dirichlet Allocation (LDA), find the top n topics of your choice in the documents and display the topics list of words and distribution including visualization.

(12 marks)

4.2. Using a package like pyLDAvis, provide a Intertopic Distance interactive Map including the most salient words per topic.

(5 marks)

4.3. Create a GitHub repository containing:

- The project documentation (Questions 1-4)
- The project datasets
- The Notebooks showcasing the flow of analytics for all questions 1 to 4
- A deployment folder with a README file containing easy deployment for the design in Question 1&2

(8 marks)

[Sub Total 25 Marks]

